



CASE HISTORY | WATER & WASTEWATER

Gujarat's Largest Industrial Water Reuse Project

20 MLD Reverse Osmosis Plant on Build-Own-Operate Model · Nandesari Industries Association, Gujarat

SITE INFORMATION

LOCATION:
Nandesari, Gujarat, India

INDUSTRY:
API & Chemical Manufacturing

APPLICATION:
Brackish Water RO, ZLD, BOO

CAPACITY:
20 MLD permeate

CLIENT:
Nandesari Industries Association (NIA)

- TDS reduced from 5,000 PPM to <200 PPM
- 95%+ system recovery
- 150 industries served via metered pipeline
- 24/7 automated SCADA operation
- COD and suspended solids reduced to nil

TECHNOLOGIES USED

- ▶ Multi-stage Brackish Water RO
- ▶ Low Fouling RO Membranes
- ▶ Pre-treatment System
- ▶ Zero Liquid Discharge (ZLD)
- ▶ PLC-controlled SCADA
- ▶ Remote Plant Monitoring
- ▶ Digital Twin (in progress)

DELIVERY MODEL

Build-Own-Operate (BOO). Permionics owns and operates the plant; treated water is metered and sold to 150 member industries through a dedicated pipeline.

OPERATIONS

Fully trained manpower, 24/7. Automated PLC-SCADA control. Plant analytics accessible remotely. Moving toward digital twin for predictive maintenance.

5,000 PPM → <200 PPM · 95% Recovery · 150 Industries · 1 Plant

INTRODUCTION

Nandesari Industrial Estate in Gujarat is home to 150 chemical and pharmaceutical production units operating under the Nandesari Industries Association (NIA). Over time, industrial wastewater intrusion into the estate's shared borewells had driven feed water TDS to 5,000 PPM, causing widespread production quality failures across member units. No individual unit had the scale or mandate to solve a shared infrastructure crisis on its own.

NIA approached Permionics to design and operate a centralised water treatment solution capable of purifying contaminated groundwater at scale and redistributing clean, process-grade water to all 150 member units. Permionics responded with Gujarat's largest industrial water reuse project — a 20 MLD reverse osmosis plant engineered, built, and operated by Permionics under a full Build-Own-Operate (BOO) model.

THE CHALLENGE

The Problem. Industrial effluent intrusion had contaminated shared groundwater across the entire estate, raising TDS to 5,000 PPM — 25 times the acceptable threshold for industrial process use (<200 PPM). Turbidity, hardness, silica, and COD were all elevated, causing quality rejections in pharmaceutical and chemical production lines. Because all 150 units drew from the same source, this was an estate-wide crisis, not an individual plant problem.

Existing Limitations. No centralised treatment infrastructure existed. The scale of contamination required a high-recovery system capable of handling variable feed quality, and any solution needed to serve the entire estate simultaneously, with guaranteed uptime and consistent permeate quality.

THE TECHNICAL SOLUTION

System Design. Permionics deployed a multi-stage brackish water RO system with pre-treatment, configured to handle a 24 MLD feed and deliver 20 MLD of purified permeate at <200 PPM TDS. The system uses Permionics' low fouling RO membranes, engineered for high-salinity, variable-quality industrial feed water.

Automation & Control. The plant is fully automated via a PLC-controlled SCADA system with remote monitoring capability. This allows Permionics engineers to track performance, trigger alarms, and adjust operating parameters without on-site intervention, ensuring 24/7 uptime and consistent permeate quality.

BOO Delivery Model. Rather than a one-time EPC handover, Permionics owns and operates the plant. Treated water is metered and sold to all 150 NIA member industries through a dedicated pipeline network, eliminating individual CAPEX for each unit and creating a reliable, shared water utility.



The 20 MLD reverse osmosis plant installed and operated by Permionics at Nandesari. The SCADA-controlled system runs 24/7 and delivers purified water to 150 industrial units via metered pipeline.

Feed Water Analysis

Parameter	Unit	Feed Value
Total Dissolved Solids	ppm	5,000
Suspended Solids	ppm	25
Total Hardness (as CaCO ₃)	ppm	<2,000
Silica (as SiO ₂)	ppm	<44
COD	ppm	<100
Turbidity	NTU	<10
Colour	—	Tinge Yellow
pH (at 32°C)	—	7.11



Partnership agreement between Permionics and the Nandesari Industries Association (NIA), Gujarat — enabling Gujarat's largest industrial water reuse project under a Build-Own-Operate model.

PERFORMANCE STABILITY

- ▶ 95%+ water recovery, sustained continuously
- ▶ SCADA automation maintains consistent outlet quality despite feed variability
- ▶ Remote monitoring enables real-time fault detection without on-site dependency
- ▶ Plant moving toward digital twin for predictive maintenance

VALUE DELIVERED

- ▶ 96% TDS reduction (5,000 PPM to <200 PPM)
- ▶ COD, suspended solids, turbidity reduced to nil
- ▶ 150 units freed from individual treatment CAPEX
- ▶ Process-grade water recycled directly back into production
- ▶ Closed-loop supply eliminates untreated discharge to water bodies

ENVIRONMENTAL IMPACT

No untreated industrial effluent discharged to groundwater or surface bodies. Purified water returned to process, completing the loop. Supports member industries' ZLD compliance objectives.

FOR MORE INFORMATION

Permionics Membranes Pvt. Ltd.
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Water & Wastewater Division
India · Asia-Pacific · Global

OPERATIONAL RESULTS

96% TDS REDUCTION 5,000 → <200 PPM	95%+ WATER RECOVERY SUSTAINED CONTINUOUSLY	150 INDUSTRIES SERVED VIA METERED PIPELINE	20 MLD PERMEATE CAPACITY GUJARAT'S LARGEST
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Before & After: Water Quality Comparison

Parameter	Unit	Feed (Raw)	Treated Water
Feed Flow Rate	MLD	24	20
Total Dissolved Solids	ppm	5,000	<200
Suspended Solids	ppm	25	Nil
Total Hardness (as CaCO ₃)	ppm	<2,000	<50
Silica (as SiO ₂)	ppm	<44	<2
COD	ppm	<100	Nil
Turbidity	NTU	<10	Nil
Colour	—	Tinge Yellow	Transparent White
pH (at 32°C)	—	7.11	6.5 – 7.0



Feed water (left, orange-tinted, 5,000 PPM TDS) vs. Permionics-treated permeate (right, transparent, <200 PPM TDS). Visual proof of the system's performance.

FIGURE 1: VISUAL TRANSFORMATION

The before-and-after comparison captures the scale of the challenge and the precision of the solution. Feed water drawn from contaminated borewells was visually turbid, tinge-yellow, and laden with dissolved salts, hardness, silica, and organic load. The Permionics-treated permeate is fully transparent, with TDS below 200 PPM and suspended solids, COD, and turbidity reduced to nil.

This quality of output, delivered reliably at 20 MLD, enables the 150 member industries of NIA to use recycled treated water directly in their production processes, eliminating freshwater dependence and supporting ZLD compliance.

CONCLUSIONS

Gujarat's largest industrial water reuse project demonstrates what centralised membrane infrastructure can achieve at estate scale. By combining high-recovery RO technology with a Build-Own-Operate delivery model, Permionics converted a shared groundwater crisis into a self-sustaining water utility — delivering process-grade water to 150 industries from a single automated plant running continuously at 95%+ recovery. Over the project's life, this translates to millions of cubic metres of contaminated water treated and recycled, zero industrial effluent discharged untreated, and an entire industrial estate freed from individual water treatment burden. Permionics has to date recycled over 10 million m³ of wastewater globally, helping 10,000+ industries achieve their sustainability goals. The Nandesari project stands as proof that scale, reliability, and sustainability are not trade-offs — they are the same solution, engineered correctly.